
Agent-to-Agent (A2A) Protocols and Production Multi-Agent Deployments

Competitive Landscape Assessment | May 2026

Classification: Complimentary research for BAIA members

Version: 1.0 | 15 May 2026

Next scheduled update: Q3 2026

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Methodology

Research conducted 15 May 2026 using multi-source web intelligence across vendor documentation, Linux Foundation publications, arXiv preprints, SEC filings, conference proceedings, analyst reports (Gartner, Everest Group), press releases, and standards body publications (NIST, FIDO Alliance, OpenID Foundation). All production claims are classified as Verified (named customer + public case study or regulatory filing), Vendor-Claimed (vendor states production use without named customer or independent audit), or Pilot/Preview (explicitly labelled as beta, preview, or pilot by the vendor).

1. Executive Summary

The A2A protocol landscape consolidated decisively in 2025-2026 around the Linux Foundation's Agent2Agent project (v1.0 stable spec announced April 9, 2026; 150+ organisations; 22,000+ GitHub stars). IBM's ACP merged into A2A in September 2025, and Cisco's AGNTCY project co-located under the same foundation. As of May 2026, no publicly-named cross-vendor A2A deployment combines a named end-customer with quantified scale evidence.

"Production" today is dominated by single-vendor multi-agent stories: Salesforce Agentforce at \$800M ARR in Q4 FY26, Microsoft "Ask Microsoft" serving millions with 70% fewer human escalations, AWS Bedrock AgentCore at Epsilon/Cox Automotive/Ericsson, and Anthropic Managed Agents at Notion/Rakuten/Sentry/Asana.

Three Biggest Governance/Security Gaps

- 1. No standardised agent identity or delegation-chain primitive. OAuth tokens lack enforced expiry and OIDC-A remains an arXiv draft, not a ratified standard.**
- 2. No federated audit/observability at the inter-agent boundary. Cross-vendor tracing is entirely implementer-built work.**
- 3. Undefined liability allocation for autonomous agent decisions. No legal framework or protocol-level mechanism exists.**

Timeline Assessment

Mainstream cross-vendor production A2A is 18-30 months away (late-2027 to mid-2028). Gartner's April 2026 Hype Cycle for Agentic AI places AI agent development platforms at the Peak of Inflated Expectations with a 2-5 year time-to-mainstream rating. Separately, Gartner predicts over 40% of agentic AI projects will be cancelled by end of 2027 due to escalating costs, unclear value, or inadequate risk controls.

Market Maturity

Protocol-and-framework layer: early production. Cross-vendor deployment layer: late pilot. Governance/identity layer: pre-standard.

2. Open Standards Landscape

2.1 Agent2Agent (A2A) Protocol

Field	Detail
Governing body	Linux Foundation (LF AI & Data); seeded by Google Cloud June 2025
Version	v1.0 stable (April 9, 2026); transport HTTP + SSE + JSON-RPC 2.0; Apache 2.0
Adoption	150+ organisations; 22,000+ GitHub stars; 5 SDKs (Python, JS, Java, Go, .NET)
TSC Members	Google, Microsoft, AWS, IBM, Cisco, Salesforce, ServiceNow, SAP
Key design	Agent Card discovery (.well-known/agent-card.json); stateful task lifecycle; client-remote model; transport bindings for JSON-RPC, gRPC, REST
v1.0 changes	Signed Agent Cards (JWS); multi-tenancy; multi-protocol bindings; version negotiation
Gaps	No enforced token lifetime; identity delegated to OAuth; Agent Card signing optional; no built-in audit envelope; no payment primitives; session smuggling risk

2.2 IBM ACP - Merged into A2A

IBM's Agent Communication Protocol merged into A2A under the Linux Foundation in August 2025. The ACP team now contributes to A2A; BeeAI migrated via the A2AServer adapter. Do not build to ACP for new deployments.

2.3 Agent Network Protocol (ANP)

Open-source alternative pursuing a fully decentralised "Internet of Agents" using W3C DIDs. Three-layer architecture: Identity (DID:WBA), Meta-Protocol (natural-language negotiation), Application (JSON-LD semantic capability descriptions). Technically compelling trust model but no hyperscaler backing. Production readiness is 2-3 years away.

2.4 Agent Payments Protocol (AP2)

Donated by Google to the FIDO Alliance on April 28, 2026. Adds payment authorisation, receipts, and DID-based key distribution. 60+ organisations including Mastercard, Visa, PayPal, American Express, Coinbase, Salesforce, Shopify. The Agentic Authentication Technical Working Group is co-chaired by CVS Health, Google, and OpenAI. Payments TWG led by Mastercard and Visa.

2.5 OpenID Connect for Agents (OIDC-A)

Proposed extension (arXiv 2509.25974). New ID-Token claims for agent type/model/provider, delegation chain validation, attestation evidence, capability-based authorisation. Not yet ratified by the OpenID Foundation. No production deployments.

2.6 AGNTCY Project (Cisco / Linux Foundation)

Donated to the Linux Foundation July 2025; 75+ supporting companies; formative members include Cisco, Dell Technologies, Google Cloud, Oracle, Red Hat. Key components: Agent Directory (DNS-equivalent for agents), Agent Identity (task-based lifecycle), SLIM (quantum-safe MLS-encrypted gRPC messaging), Observability SDK. Interoperable with A2A Agent Cards and MCP server descriptions.

2.7 Feature Coverage Comparison

Capability	A2A v1.0	MCP	ANP	AGNTCY	AP2	OIDC-A
Agent-to-agent transport	Yes	No	Yes	Yes (SLIM)	No	No
Agent-to-tool transport	Partial	Yes	Partial	Via Gateway	No	No
Capability discovery	Yes (Card)	Yes	Yes	Yes (Directory)	N/A	No
Decentralised identity	No (OAuth)	No	Yes (W3C DID)	Partial	Yes (VCs)	Proposed
Payment primitives	No	No	Via AP2	No	Yes	No
Audit/observability	Implementer	Implementer	Implementer	Yes (OASF)	Yes (signed)	No
Quantum-safe transport	No	No	No	Yes (SLIM)	No	No
Cross-vendor adoption	Very broad	Very broad	Small	Medium	Broad (payments)	None

3. Platform Vendor Analysis

3.1 Google Cloud (Vertex AI / ADK)

A2A originator. Native in ADK and Vertex AI Agent Engine. Hub-and-spoke architecture. Production evidence: Deloitte's 100+ ready-to-deploy agents on ServiceNow + Google Cloud A2A - no named end-customer with metrics. OAuth 2.0 and Agent Card discovery. Lock-in risk: medium.

3.2 Microsoft (Copilot Studio + Agent Framework + Azure AI Foundry)

A2A generally available in Copilot Studio as of April 2026. Supports first-party Foundry agents, Fabric Data agents, M365 SDK agents, and third-party A2A agents. Agent Framework 1.0 GA (April 2026) for .NET and Python.

Production evidence: "Ask Microsoft" web agent serving millions of customers on microsoft.com with 61% lower latency, 70% fewer human escalations, and customers 10x more likely to sign up. Architecture is intra-vendor sub-agents (Azure, M365, pricing, trials). Coca-Cola Beverages Africa saves planners 1-1.5 hours daily.

Governance: Immersive Prompt Builder (GA), evaluation automation APIs, Analytics Viewer role, content moderation settings. AI Agent Builder Associate certification launched. Lock-in risk: medium-high.

3.3 Salesforce (Agentforce / Agentforce 360)

Foundational A2A project member. Contributed the Agent Card concept. Agentforce 360 ecosystem opened to ISVs in Q4 FY26. Agent Fabric for multi-vendor coordination; Agent Broker beta April 2026, GA June 2026.

Production evidence (largest verified single-vendor scale): \$800M ARR in Q4 FY26 (+169% YoY); 29,000+ cumulative deals; 18,500+ customers across 124 countries; 1.5M+ support requests handled. Named: Finnair (targeting 80% case resolution), UChicago Medicine (2.5M patient inquiries/year), Nexo (62% resolution), 180+ organisations on Agentforce IT Service in 4 months.

Governance: Trust Layer (PII privacy, hallucination mitigation, bias controls), Agentforce Command Center, deterministic execution, Trusted Agent Identity, LLM Governance on AI Gateway. Lock-in risk: high (CRM data gravity).

3.4 AWS (Bedrock AgentCore + Strands Agents)

A2A server support in AgentCore Runtime GA November 2025. Containerised A2A servers with 8-hour sessions. Multi-framework: Strands, LangChain, LangGraph, Google ADK, OpenAI Agents SDK, Claude Agent SDK.

Production evidence: Epsilon (20+ agents, 200M+ consumer profiles, 30% campaign setup reduction), Cox Automotive (17 enterprise solutions in <1 year), Ericsson (double-digit gains across tens of thousands of workforce). AgentCore SDK downloaded 1M+ times.

Governance: AgentCore Identity (Cognito), Gateway (MCP interface), Observability (OpenTelemetry + CloudWatch + X-Ray), Agent Registry (centralised cross-org governance), VPC connectivity. Lock-in risk: medium.

3.5 Anthropic (Claude Managed Agents)

No native A2A. Single-vendor Claude infrastructure. Launched April 8, 2026 in public beta. Sub-agents and Agent Teams are intra-Claude only. Multi-agent coordination remains in research preview.

Production evidence: Notion (dozens of parallel tasks per session), Rakuten (specialist agents live in under a week), Sentry (root cause to merged PR in weeks instead of months), Asana, Atlassian, Allianz.

Governance: Sandboxed code execution, OAuth flows, credential vault, session checkpointing. No A2A interop; high lock-in. Pricing: standard Claude token rates + \$0.08 per session-hour.

3.6 Other Vendors

ServiceNow: A2A client/server support in Zurich Patch 4 (December 2025). Joint announcement with Google Cloud April 2026 for Autonomous Network/IT Operations - currently in preview, GA targeted H2 2026.

SAP: Joule bi-directional A2A scheduled GA in Q4 2026. Named customer LC Waikiki (70% efficiency increase, 50% error reduction) is single-vendor Joule.

Workday: Agent Gateway with A2A; Agent System of Record (ASOR). 15+ partner network including Accenture, AWS, Google Cloud, IBM, Microsoft. Platform processes 1 trillion+ transactions/year across 11,000+ organisations.

4. Framework-Level Adoption

Framework	A2A Support	MCP Support	Production Maturity	Notes
Google ADK	Native (created it)	Yes	Medium-high	Optimised for Gemini; A2A first-class
LangGraph	Via integrations	Yes (native)	Highest	v1.0 GA Oct 2025; LangSmith tracing
CrewAI	Added (2026)	Yes	Medium-high	Role-based; fastest prototyping; 44,600+ stars
Microsoft Agent Fwk	Native (1.0 GA)	Yes	Medium	Replaced AutoGen + Semantic Kernel
Strands (AWS)	Native	Yes	Medium-high	First-class on AgentCore
OpenAgents	Native (both)	Yes (native)	Low (young)	Only framework with native MCP + A2A
OpenAI Agents SDK	No	Yes (April 2026)	Medium	Proprietary handoff model; OpenAI-locked
Claude Agent SDK	No	Yes (native)	High (single-vendor)	Powers Claude Code; no A2A
AutoGen / AG2	No	Partial	Maintenance	Shifted to MS Agent Framework; avoid for new work
BeeAI (IBM)	Via adapter	Yes	Low-medium	Migrated from ACP; Linux Foundation

5. Governance and Security Gap Analysis

5.1 Protocol-Level Gaps

Gap	Severity	Affected Protocols	Source
No enforced token lifetime (long-lived OAuth tokens)	Critical	A2A, MCP	arXiv 2505.12490; CVE-2025-1198; CVE-2025-1801
Agent Card signing optional, not mandatory	High	A2A v0.3+	Semgrep security guide
No standardised delegation chain for agent identity	Critical	All (except OIDC-A draft, ANP)	arXiv 2509.25974; OpenID Foundation April 2025
Session-smuggling / prompt-injection across stateful sessions	High	A2A	InfoQ November 2025 analysis
No protocol-level consent orchestration	High	A2A, MCP	arXiv 2505.12490
Cross-protocol vulnerability bleed (A2A auth flaw to MCP access)	High	A2A + MCP combined	Alcaraz, Medium April 2025
No mandatory audit envelope for cross-agent messages	Medium-high	All	CSA MAESTRO framework
No quantum-safe transport mandate	Medium	All except AGNTCY SLIM	Cisco AGNTCY

5.2 Standards Body Activity

Body	Initiative	Status	Expected Deliverable	Timeline
Linux Foundation	A2A Project	Active	v1.x releases; AGNTCY stack	Continuous
FIDO Alliance	Agentic Auth TWG + Payments TWG	Launched April 2026	Agent-auth standard; AP2 into financial switching	H2 2026-2027
OpenID Foundation	Identity for Agentic AI	Pre-standard draft	OIDC-A 1.0 ratification	2026-2027
NIST CAISI	AI Agent Standards Initiative	Launched Feb 2026	Interoperability and security guidance	Late 2026-2027
NIST COSAIS	SP 800-53 Agent Overlays	In development	Single-agent + multi-agent control overlays	Draft 2026; final 2027

NIST NCCoE	Agent Identity & Auth Concept Paper	Pre-project (Feb 2026)	Reference architecture (OAuth + SPIFFE + MCP)	2026-2027
MITRE ATLAS	Agentic kill-chain expansion	In development	Multi-agent lateral movement tactics	2026

6. Production Deployment Evidence

Deployments are classified as Verified (named customer + public case study or regulatory filing), Vendor-Claimed (vendor states production without named customer or independent audit), or Pilot/Preview (explicitly labelled as such).

Organisation	Platform	Scale Evidence	Classification
Microsoft (Ask Microsoft)	Copilot Studio sub-agents	Millions served; 61% lower latency; 70% fewer escalations	Verified, intra-vendor
Salesforce (platform total)	Agentforce	\$800M ARR Q4 FY26; 29,000 deals; 18,500+ orgs	Verified, intra-vendor
UChicago Medicine	Agentforce	2.5M patient inquiries/year	Verified, single-vendor
Epsilon (Publicis)	AWS AgentCore + Strands	20+ agents; 200M+ profiles; 30% setup reduction	Verified, single-vendor
Cox Automotive	AWS AgentCore + Strands	17 solutions in <1yr; 3 products to prod in 5 weeks	Verified, single-vendor
Ericsson	AWS AgentCore	Double-digit gains across tens of thousands workforce	Vendor-claimed
Notion	Anthropic Managed Agents	Dozens of parallel tasks per session	Vendor-claimed at launch
Rakuten	Anthropic Managed Agents	Specialist agents live in under a week	Vendor-claimed at launch
Sentry	Anthropic Managed Agents	Root cause to merged PR in weeks vs months	Vendor-claimed at launch
ServiceNow + Google Cloud	A2A + A2UI + MCP	Autonomous Ops solutions	Pilot/Preview
Deloitte (100+ agents)	ServiceNow + Google Cloud	Cross-vendor A2A	Vendor-claimed capability

Key finding: Production scale is overwhelmingly single-vendor multi-agent. Cross-vendor A2A is still pilot/preview. No publicly-named deployment combines a named end-customer, confirmed cross-vendor A2A use, and quantified scale.

7. Timeline Assessment

Milestone	Estimated Date	Confidence	Dependencies
A2A v1.0 stable spec	Done (April 2026)	Confirmed	-
Microsoft Copilot Studio A2A GA	Done (April 2026)	Confirmed	-
AWS AgentCore A2A GA in Runtime	Done (Nov 2025)	Confirmed	-
Anthropic Managed Agents beta	Done (April 2026)	Confirmed	-
Google donates AP2 to FIDO Alliance	Done (April 2026)	Confirmed	-
SAP Joule bi-directional A2A GA	Q4 2026	High	SAP Sapphire announcement
ServiceNow Autonomous Ops GA	H2 2026	Medium-high	ServiceNow + Google roadmap
First cross-vendor A2A reference customer	H1 2027	Medium	Vendor PR cycle + customer willingness
NIST agent-specific SP 800-53 overlays	Draft late 2026; final 2027	Medium	NIST process
OIDC-A or equivalent ratified	2027	Medium-low	OpenID Foundation pace
Mainstream cross-vendor A2A	2028-2029	Medium	Identity, audit, liability standards
40%+ of agentic AI projects cancelled	By end 2027	Medium-high	Cost/value reckoning (Gartner)
40% of enterprise apps with AI agents	End 2026	High	Gartner (already partly realised)

8. Recommendations

Standards to Track

1. Linux Foundation A2A - v1.x point releases, especially mandatory Agent Card signing and audit envelopes. Primary source: github.com/a2aproject.
2. FIDO Alliance Agentic Auth TWG + Payments TWG - likely convergence point for cross-vendor agent identity and AP2 integration into financial switching.
3. NIST COSAIS multi-agent overlay + NCCoE agent identity reference architecture - federal and regulated-industry procurement will follow these.
4. OpenID Connect for Agents (OIDC-A) - only existing OAuth-compatible draft addressing delegation chains.

Vendor Moves to Monitor

Microsoft Agent Framework + Foundry A2A maturation. Salesforce Agentforce 360 cross-vendor reference customers. AWS AgentCore A2A across all services (currently Runtime only). Anthropic multi-agent coordination exiting research preview. ServiceNow + Google Cloud Autonomous Operations GA. OpenAI's A2A position (conspicuously absent from A2A working group).

Research Papers to Read

arXiv 2504.16902 - Building A Secure Agentic AI Application Leveraging A2A Protocol (MAESTRO threat model).

arXiv 2505.12490 - Improving Google A2A Protocol (token-lifetime gaps, prompt-injection benchmarks).

arXiv 2509.25974 - OpenID Connect for Agents (OIDC-A) 1.0.

arXiv 2508.00007 - Agent Network Protocol Technical White Paper.

CSA Agentic AI Red Teaming Guide and MAESTRO framework documentation.

NIST NCCoE concept paper on AI Agent Identity and Authorization (February 2026).

Key Conferences (Next 12 Months)

Linux Foundation Open Source Summit. AWS re:Invent (Nov-Dec). Microsoft Ignite. Google Cloud Next (April). Salesforce Dreamforce / Agentforce World Tour. Anthropic Code with Claude. SAP Sapphire (May). RSA Conference / Black Hat (A2A/MCP security tracks). OpenID Foundation workshops.

Communities to Engage

a2aproject GitHub organisation and Linux Foundation A2A mailing list. AGNTCY collective (agntcy.org). Cloud Security Alliance MAESTRO working group. FIDO Alliance Agentic Authentication TWG. OpenID Foundation Identity for Agentic AI working group.

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About SolNet

SolNet specialises in AI agent security, governance, and competitive intelligence for the UK AI ecosystem. We help organisations understand the rapidly evolving agent landscape and build secure, governed AI systems.

About the Author

John (Sol) Soulsby, Founder

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